CS 4501 · ALPHA

CLEANA-SAURUS

From ”clean my Downloads” to **freeing real space on developers’ laptops**, safely.

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01 · ACTIVITY 1 · WHAT WE TESTED

Will students let an AI clean their files if every plan is checked first?

Who

10 testers on their own laptops: 5 CS students, 5 non-CS (4 family, 1 student).

Test 1 · watched

20-minute think-aloud: free up space, organize files, scan for malware.

Test 2 · alone

3 days unsupervised, measured with the app's opt-in usage log (8 of 10 sent it).

Change from plan: 15 testers planned, 10 ran. Treat percentages as directional.

02 · RESULTS

They trusted it. It didn't solve their problem.

KPI	TARGET	RESULT	
Ran a plan in session 1	≥ 60%	**70%**	MET
Unsafe actions reaching a user	0	**0**	MET
Actions undone	≤ 10%	**11%**	CLOSE
Used again within 3 days	≥ 40%	**30%**	CLOSE
Space freed vs. space wanted	≥ 50%	**~2%**	MISSED

And the FREED counter said 12.4 GB while **0 bytes** left the disk: the Dino Bin sat on the same drive.

03 · DECISION: PIVOT

Keep the trust engine. Aim it where the space is.

38 GB

median **rebuildable dev clutter** per CS tester

1.4 GB

median duplicates we could free

- **5 of 5** CS testers named a place the app wouldn't go: `node_modules`, Python envs, Xcode, Docker, model caches.
- Trust carried over: CS testers met **all 3** trust KPIs.
- Dev clutter grows back every class and hackathon, which fixes repeat use.
- Rivals do one ecosystem each (npkill, DevCleaner, `docker prune`) and none prove a folder can be rebuilt.

04 · THE PIVOT

Cleanasaurus for developers' laptops, starting with CS students

I'm out of space. I'm done with CS 3140 and the hackathon, but don't touch my capstone.

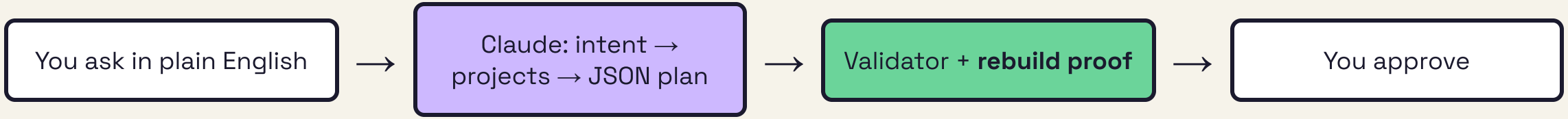
I found 38 GB I can **prove** you can rebuild: 6 `node_modules` (npm ci), 3 Python envs, old Xcode data, unused Docker images. I left `hoohacks-2026` alone: it has uncommitted work.

New hypothesis

CS students over 80% full will, without us in the room, approve a validated plan that frees **≥10 GB** in session 1 (≥60%), and ≥40% run it again within 30 days.

Customer-segment + zoom-in pivot. The vision is unchanged.

Claude writes the plan. Plain code decides if it can run.



CORE REQUIREMENT	SETS US APART	MVP	LLM
FR1 Plain English → plan		✓	✓
FR2 Validator gate · FR3 self-correction (3 tries)	✓	✓	✓
FR4 Ask instead of guessing, and remember		✓	✓
FR6 Rebuild proof · FR7 "done with CS 3140" → projects	✓		✓
FR11 Malware only from scanners · FR15 zones · FR17 preview		✓	

06 · STORIES → GITHUB ISSUES

29 user stories, every one a tracked issue

> **US-05** · As a dev student, I want Cleanasaurus to delete a folder only when it can prove I can rebuild it, so that freeing space never costs me work.

- **Given** a gitignored folder with its lockfile, no secrets, in a project unchanged 30 days, **then** it passes and the rebuild command is saved.
- **Given** uncommitted changes, **then** nothing in it is offered, and the dino says why.
- **Given** git tracks a file inside, or no lockfile, **then** the validator rejects it and tells the planner.

PRIORITY	STORIES	POINTS
Built in MVP (closed)	US-24–29	0
Must (Alpha)	22 stories	78
Should	US-13, 22	5
Could	US-17, 20, 23	16
Issues #2–#30 · labels: priority, alpha, status · 4-sprint plan		

07 · ACTIVITY 3 · TEST-DRIVEN DEVELOPMENT

Tests for every issue first. Watched them fail. Then built.

c1c337d
Tests for all 23 open issues, no code
11 failing · 46 todo

d2f885d
Rebuild proof, labels, JS + Python
US-02, 06, 07 green

454d384
Validator, real deletes, measured FREED
US-05, 11, 12 green

14138e7
Regression test first, then fix
REG-2 red → green

62 tests passing (was 25)

0 failing

46 todo: written, story not built

14/14 end-to-end UI steps

CI runs npm test on every push · tests per issue in app/test/stories/ · red and green runs saved in tdd/

Claude drafted the tests. We checked them, and they caught real bugs.

WHAT WE FOUND	TYPE	WHAT WE DID
A git repo at a home folder would make every project look ”not gitignored”	Missing edge case	Test added **before** code; proof ignores home-level repos
Only happy paths: no lockfile, nested node_modules, pnpm hard links, renamed venv	Missing edge cases	Added 6 edge-case tests
US-11 test assumed one failure; deleting package.json also dirties the repo	Wrong assumption	Test checks for the right condition among all failures
Planner re-proposed identical source files in code projects, failing all 3 tries	**Real bug** (REG-1)	Planner skips NEVER zones, drops rejected actions
Dino asked about ”6 versions of index.js”	**Real bug** (REG-2)	Failing test first, then fix

09 · THE ALPHA RUNNING



6 folders proposed → validator rejected 3 (uncommitted work, active 4 days ago, a `.env` secret) → try 2 passed: 3 folders, each with its rebuild command. ~150 MB measured freed on disk.

What's left before the pivot test

1. Wire package caches and conda export into plans
2. US-01 storage breakdown → US-03 ”done with CS 3140”
3. US-15 one-click rebuild · US-14 undo one action
4. US-18 one-line summary · US-19 unsigned ≠ malware
5. Xcode, Docker, ML caches · storage watch · Go Pro fake door

Risks we’re watching

- Developers expect free tools: test willingness to pay with the Go Pro fake door
- Each ecosystem needs its own rule: five cover 91% of what we measured
- A wrong delete before a deadline is worse than a lost PDF: anything unproven stays NEVER

Thank you! · github.com/natekim416/cs4501_startup